

A Project Report On

FIRE AND METAL DETECTION ROBOT

Submitted in partial fulfillment of the requirement for the award of Degree of

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS & COMMUNICATION ENGINEERING

SUBMITTED BY

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CERTIFICATE

This is to certify that the project entitled

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Is record of bonafide work carried out by

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Under our supervision and that no part of this project has been submitted for the award of any other Fellowship or similar titles or prizes and that the work has not been published in scientific or popular journal or magazine.


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ABSTRACT

Specifically, in this program's working environment, the designing and construction of a robot that can detect fire and metal mines using sensor technologies and microcontroller (AT89C51) was undertaken. While earning an engineering degree, the task "Fire and metal detecting robot" was completed, demonstrating electronics engineering domain competence.

It efficiently performed robot designing and interface of the associated components with the controller using the best technologies and engineering methodologies during the task. Various embedded systems and other technologies were studied for the project designing section in order to regulate system operations with more efficiency and precision.

Fundamentally, the microcontroller has been discovered to be one of the systems capable of conducting required control operations;

nonetheless, such controllers are mostly designed to accomplish the intended duty. In addition to the planning phase, the sensor technologies examined to detect the presence of smoke/fire and metals were investigated.

This review provided a thorough comprehension of the subject, and the technique for connecting the sensor technology module with the microcontroller (AT89C51) was also discussed. All of the technical modules required for the optimum design of the suggested robot for fire and metal mine detection were extensively investigated.

According to the program design idea, the AT89C51 microcontroller was employed as the principal controller, with sensor modules for sensing metal objects and fire. It took a lot of effort to get the main work running smoothly and with the best work schedule possible. With sensor technologies and microcontrollers, it was possible to design and create a robot that could detect fire and metal mines in this development working environment.

The project design and development approach, which comprised both software and hardware configuration in terms of circuit design, circuit simulation, and hardware circuit testing analysis, was stated in the planning process.

In contrast to the LCD for showing important information and the buzzer, two motors and a motor driver IC drove the robot. A flame sensor was chosen for fire detection, and a metal detecting circuit was constructed using a 555-timer IC configured with R, L & C circuit components, however using both sensor components being interfaced to the microcontroller AT89C51 depending on its pin configuration.

The circuit was designed using the Proteus software package's circuit modeling and simulation environment. However, particular circuit components were chosen and expressed using standard symbols. The system components were interfaced with the microcontroller AT89C51 using the programming technique and procedure. Most crucially, the KEIL framework was used for programming, and the

HEX file was received once the program was compiled and put into the microcontroller.

The circuit interfacing in the Proteus software package was completed with reference to the programming of the AT89C51 microcontroller, based on which the simulation was run to validate the activities after delivering the power.

It was purposefully involved in the construction of a hardware circuit model of the robot and, for this general breadboard used for installing the circuit elements during the last phase. The breadboard was given the necessary power supply, and the robot's operations were shown.

DETAILED DESCRIPTION

Fire has devastated millions of hectares of land throughout the world. Hundreds of hundreds of lives, both human and animal, are lost every year. If a fire is not adequately contained, the consequences can be devastating. As a result, it is critical to identify fire early on to avoid large-scale damage.

This may be accomplished by deploying surveillance cameras to detect fire, finding the fire, and quickly contacting the fire department or disaster management staff to put the fire out without causing more damage. There are two types of fire detection systems: classic fire alarms and vision controller fire detection systems. Traditional fire alarms use proximity sensors within a certain distance to activate (e.g., smoke sensors).

Smoke may not be created in large quantities during the early stages of a fire. These devices will take longer to detect the fire in that

instance. These technologies also require human intervention to determine whether the fire is harmful. These technologies have the potential to generate false alerts. For real-time fire detection, vision sensor-assisted fire detection systems employ cameras.

As a result, it is more responsive than traditional smoke detectors and involves less human intervention. It can cover broad regions and verify fire without physically visiting the place. They are inexpensive and can offer information on the emissions of CO₂ fast and precisely in their early stages.

They are inexpensive and can rapidly and correctly offer information on how to decrease flammable, harmful product pollutants and greenhouse gases created by the fire itself. These environmental consequences are frequently underestimated, yet they are unavoidable in all fires.

Many home things catch fire when someone is sleeping or gone, resulting in various hazardous situations if the fire is not put out

quickly. So, as a mechanical engineer, the current project is to design and develop a system for detecting and extinguishing the fire. Researchers utilised a fundamental notion to comprehend from the perspective of novices or specialists in this discipline.

The hazardous atmosphere generated by combusting substances is one of the most severe risks of firefighting activities. Smoke, an oxygen-deficient atmosphere, high temperatures, and poisonous atmospheres are the four principal risks connected with these circumstances. Falls and structure collapse are additional fire hazards. Sensor technology has advanced tremendously across the world over the years.

Several sensors have been created for various purposes, including vehicle obstruction detection, monitoring and surveillance. Surveillance has primarily attracted many researchers among these various applications due to the proven capabilities of cameras. This study provides an overview of several fire detection systems and assesses the effectiveness of the various approaches examined.

TECHNICAL BACKGROUND

An alloy is used to make a fusible-element heat detector. The alloy transforms from a solid to liquid-like soldering when the temperature reaches the passivating point. An electrical circuit is formed, and an alarm is triggered, resulting in the release of a spring kept under stress.

A wired network of electrical conductors is isolated from one another by thermally sensitive insulation and wrapped in a protective covering in a dispersed heat detector. Because the insulator alters physical condition from a rigid solid to a molten form when exposed to extreme heat.

The twisting conductors generate an electrical circuit, and an alarm is triggered. The three types of dispersed heat sensors are electrically, optically, and sheathed thermocouples. In the event of a fire, the temperature causes various cable characteristics to alter.

The principle of operation of electromechanical and coated thermocouple kind heat sensors are variations in series resistance and temperature increase. The heat capacity of alloys is the operating concept of different metals type heat sensors. Two metals with different expansion rates are bonded together to make a single piece of metal utilised in fire alarms as a metal wire.

In optomechanical heat detectors, thermally sensitive insulation divides one or more fibre optic wires covered by an outer sheath. The strip closes the open electrical current, triggering an alert due to the heat. Smoke detection is critical for disaster and incident prevention. Given the wide range of smoke detection technologies and sensors presented, none have proven the ability to maintain a high frame rate while improving detection efficiency.

There is already significant demand for autonomous smoke detection systems that quickly respond and need little maintenance. These surveillance systems may detect smoke independently or provide early fire warning.

The sparks do not emerge in front of the camera during the first seconds after igniting in the latter instance, but the burning materials release pillars of smoke that fill more enormous volumes. Even when the source of the fire is hidden behind, an occurrence might be detected in such circumstances.

The detector aims to find an underground metal surface, compute the size of the uncovered metal, avoid impediments in its path from the front, left, and right, and discover a better and faster route for a certain navigation region. The components and methods of the metal detector robot construction are described in this section.

The robot detector requires both hardware and software components to work. These components were chosen based on several factors, including cost-effectiveness, accessibility, accessibility in the fabrication location, and programming convenience. The software tool for managing the detector required hardware components for fabrication and testing.

At the same time, the Arduino UNO microcontroller boards serve as the program's brains to bring the robot to the current location and operate it to identify metals and barriers sequentially. On the other hand, a brain is a microcontroller Arduino UNO board (ATmega328P) that gives commands and analyses the data.

The Arduino microcontroller comprises a physical configurable circuit board and application that runs on Microsoft Windows and is used to develop software programs for the physical board. A Visual C# programme is used to create the computer application.

Uno receives data from the computer, runs the sensors, receives their findings, analyses them before sending them to the desktop, runs the motors, moves the robot to the desired location, and stops the robot if it comes into contact with metal or impediments. Uno receives information from the computer, runs the sensors, receives their findings, analyses them before sending them to the computer, runs the motors and moves the robot to the desired location, and stops the robot if metal or obstructions are revealed sending an alarm to the computer.

Furthermore, the middleware between the computers and the robotic circuit is Bluetooth technologies, developed to serve as a wireless communication route between systems peripherals. This implies that some robot tasks may be completed autonomously, while others require computer commands. *Bluetooth* is a wireless link that allows computers and devices to communicate across short distances.

To move the robot, it must first get an instruction from the computer, after which the microcontroller will examine the evidence and issue

an order. If there are many paths in the mapping area, the detector can calculate the optimal route and transmit the data to a computer to be shown to make the best selection. The intelligent approach demonstrates the optimum route in the navigation region that requires mapping and skimming in terms of metals detection.

The system also chooses the path with the most elements if metals are identified in many locations. The suggested system's intelligence is centred on its capacity to identify impediments and analyse the findings to calculate the probability of robot arrival for the needed region.

When the robot reveals a metal and the forms and quantities of the number of minerals, it immediately stops. When the metal detector begins skimming in the target region, the robot activates the metal detector's sensors. If the detector detects a metal, the robot will halt and transmit an alarm to the computer, which will then show the received materials testing alert.

DATA TESTING

According to the set goals, this task-working environment was used to design and create a robot that can recognize fire and metal mines using sensor technologies and a microcontroller (AT89C51). The program design and development approach, which comprised both software and hardware configuration in terms of circuit design, circuit simulation, and hardware circuit testing analysis, was stated in the planning process. With the use of several functional blocks the block representation given in Fig. 1 below-

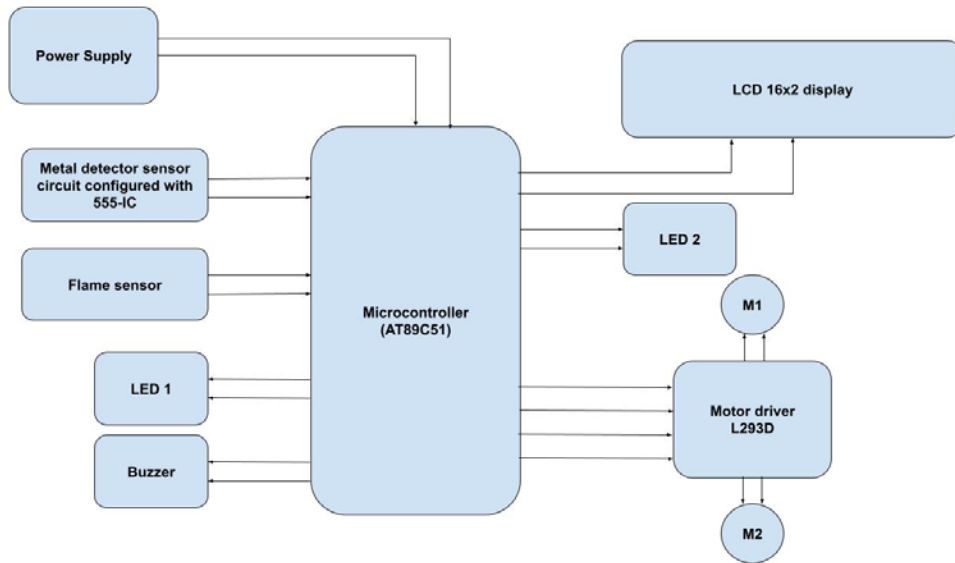


Fig. 1: Block representation

A flame sensor was chosen for fire detection, and a metal detecting circuit was created, particularly utilizing a 555-timer IC connected with R-L-C components, and both of these sensor modules were interfaced to the microcontroller AT89C51 based on its pin-configuration.

A buzzer was also included and an LCD for showing important information. However, two motors and a motor driver IC were incorporated into the microcontroller to move the robot. A time delay in the system software was also included for reset and resumption of activities with the microcontroller whereas the programming

algorithm prepared based on a process flow diagram i.e. shown below-

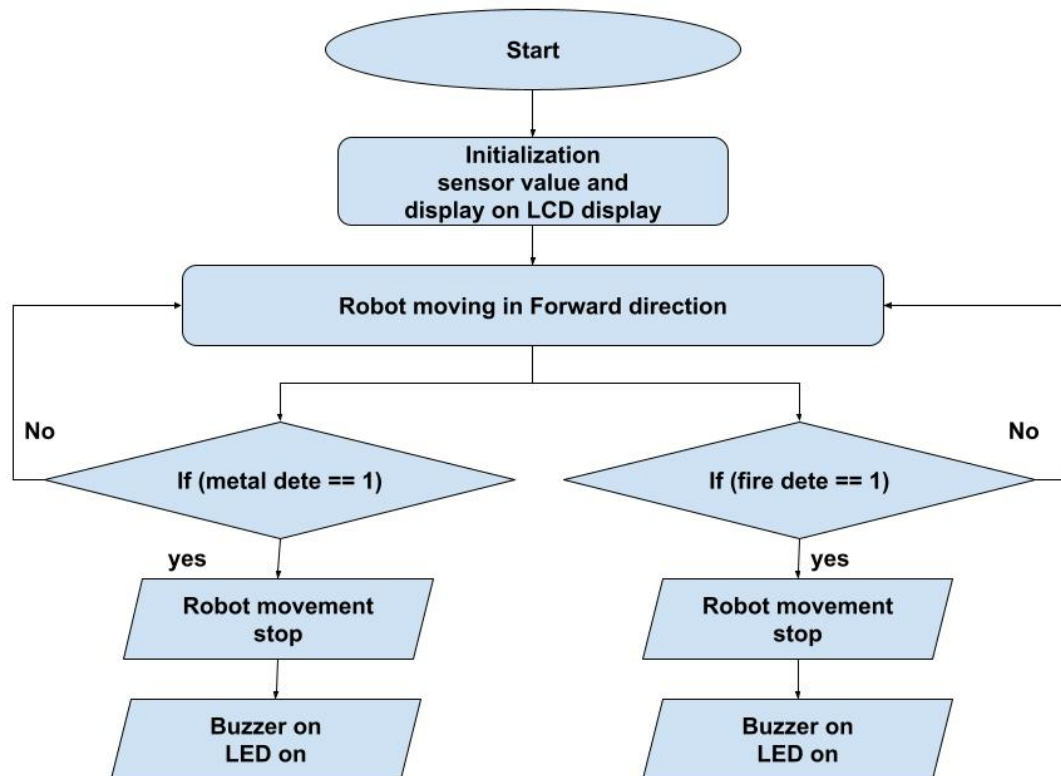


Fig. 2: Operational flow chart

According to the process flow diagram and using embedded C language the programming part of the microcontroller (AT89C51) executed in the integrated development environment of the microcontroller i.e. KEIL framework. Most crucially, a conventional IDE was used for programming, and the HEX file was received once

the program was compiled and put into the microcontroller whereas the programming procedure depicted as follows-

```
void forward();
void stop();

int rec;

char abc[16];

void L_CMD (char L_value)
{
    L_RS_PIN = 0;
    L_EN_PIN= 1;

    L_DATA_P= L_value;

    L_RS_PIN = 0;
    L_EN_PIN = 0;
}

void fun_delay (unsigned int a )
{
    char k; char l;

    for (k=0;k<a;k++)
    {
```

Fig. 3: Programming procedure

```

void fun_delay (unsigned int a )
{
    char k; char l;

    for (k=0;k<a;k++)
    {

        for (l=0;l<10;l++)
        {

        }

    }
}

void LCD_begin (void)
{
    L_CMD(0x30);
    fun_delay(30);

    L_CMD(0x38);
    fun_delay(30);

    L_CMD(0x0c);
    fun_delay(30);
}

```

Fig. 4: Programming procedure

```

void LCD_begin (void)
{
    L_CMD(0x30);
    fun_delay(30);

    L_CMD(0x38);
    fun_delay(30);

    L_CMD(0x0c);
    fun_delay(30);

    L_CMD(0x01);
    fun_delay(30);

    L_CMD(0x06);
    fun_delay(30);
}

void DIS_DATA(int L_value)
{
    L_RS_PIN=1;

    L_EN_PIN = 1;    //Strobe the enable pin

    L_DATA_P= L_value;    //Put the value on the pins

    L_RS_PIN = 1;
}

```

Fig. 5: Programming procedure

```

void DIS_DATA(int L_value)
{
    L_RS_PIN=1;

    L_EN_PIN = 1;    //Strobe the enable pin

    L_DATA_P= L_value;    //Put the value on the pins

    L_RS_PIN = 1;
    L_EN_PIN = 0;
}

void disp_fun (unsigned char curSor, unsigned char *point)
{
    L_CMD(0x80 | curSor);
    fun_delay(1);
    while(*point)
    {
        DIS_DATA(*point++);
    }
}

int ser_rec(void)
{
    while(!RI);
}

```

Fig. 6: Programming procedure

```

    L_DATA_P= L_value;    //Put the value on the pins

    L_RS_PIN = 1;
    L_EN_PIN = 0;
}

void disp_fun (unsigned char curSor, unsigned char *point)
{
    L_CMD(0x80 | curSor);
    fun_delay(1);
    while(*point)
    {
        DIS_DATA(*point++);
    }
}

int ser_rec(void)
{
    while(!RI);
    RI=0;
    return SBUF;
}

```

Fig. 7: Programming procedure

```

LCD_begin();

disp_fun(0x00," Fire and Metal ");
disp_fun(0xc0,"Detection System");
fun_delay(2000);

    while(1)
    {
        forward();
    if (fire_sen)
    {
        stop();
        buzz = 1;
        disp_fun(0x00,"Metal detected !");
        disp_fun(0xc0,"  Buzzer ON  ");
        fun_delay(2000);
        disp_fun(0x00,"Robot direction ");
        disp_fun(0xc0,"      Stop !      ");
        fun_delay(2000);
    }

    if (met1_sen)
    {
        stop();
        buzz = 1;

```

Fig. 8: Programming procedure

```

    if (fire_sen)
    {
        stop();
        buzz = 1;
        disp_fun(0x00,"Metal detected !");
        disp_fun(0xc0,"  Buzzer ON  ");
        fun_delay(2000);
        disp_fun(0x00,"Robot direction ");
        disp_fun(0xc0,"      Stop !      ");
        fun_delay(2000);
    }

    if (met1_sen)
    {
        stop();
        buzz = 1;
        disp_fun(0x00," Fire detected !");
        disp_fun(0xc0,"  Buzzer ON  ");
        fun_delay(2000);
        disp_fun(0x00,"Robot direction ");
        disp_fun(0xc0,"      Stop !      ");
        fun_delay(2000);
    }

```

Fig. 9: Programming procedure

```

        disp_fun(0x00," Fire detected !");
        disp_fun(0xc0,"  Buzzer ON  ");
        fun_delay(2000);
        disp_fun(0x00,"Robot direction ");
        disp_fun(0xc0,"      Stop !      ");
        fun_delay(2000);
    }

    else
    {
        buzz = 0;
    }
}

void forward()
{
    disp_fun(0x00," Fire and Metal ");
    disp_fun(0xc0,"Detection System");
    fun_delay(2000);
    moto1 = 1;
    moto2 = 0;
    moto3 = 1;
    moto4 = 0;
}

```

Fig. 10: Programming procedure

Overall, an AT89C51 microcontroller was used as the key component for overall system control, with a voltage regulator IC-based power supply circuit also being considered in the design process. The circuit was designed using the Proteus software package's circuit modeling and simulation environment; however, particular network elements were chosen and expressed using standard symbols.

The Proteus software program is used. It was performed in the circuit designing phase, where a flame sensor was chosen for fire detection,

and a metal detecting circuit was developed using a 555-timer IC connected with R-L-C components, but both of these sensor modules were interfaced to the AT89C51 microcontroller depending on its pin-configuration.

Aside from that, the LCD was connected to the microcontroller to show system status, and a preset was used to mimic the behaviour of a variable voltage device. The preset would change when the voltage changed, and the LCD contrast could be modified. The potentiometer was made up of a variable resistor.

The buzzer and motor driver IC were also combined with the microcontroller to allow the robot to move in different directions. The network elements were interfaced with the microcontroller AT89C51 using the programming technique and procedure.

The circuit interface was completed in the Proteus software package with reference to the programming of the AT89C51 microcontroller, and the simulation was run to test the operations after giving power.

As a result, the circuit design and simulation analysis performed in Proteus are depicted as follows:

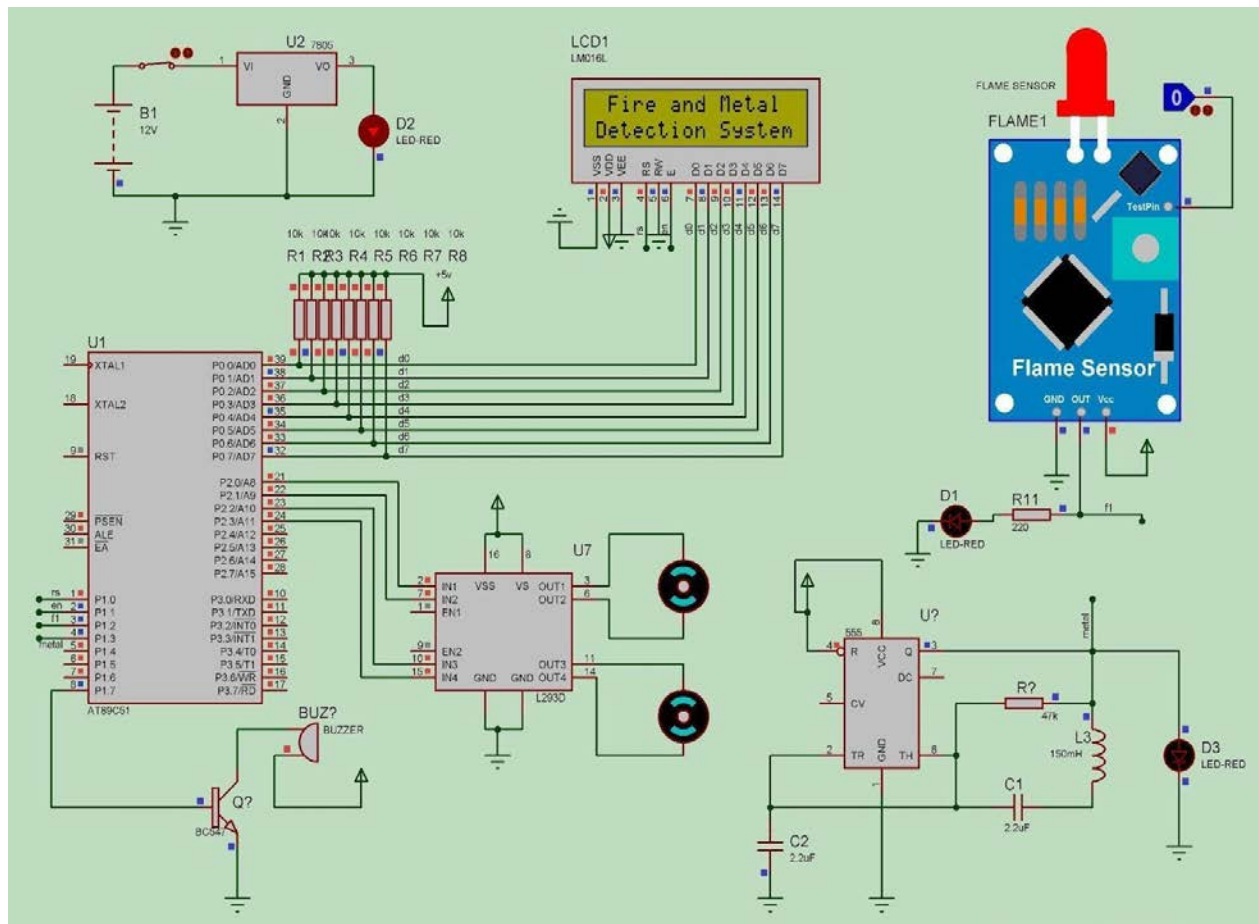


Fig. 11: Circuit simulation procedure

Under the simulation need power given to the circuit and then initial information displayed on LCD as given in Fig. 11 and then robotic vehicle starts movement in forward direction as shown in below Fig.-

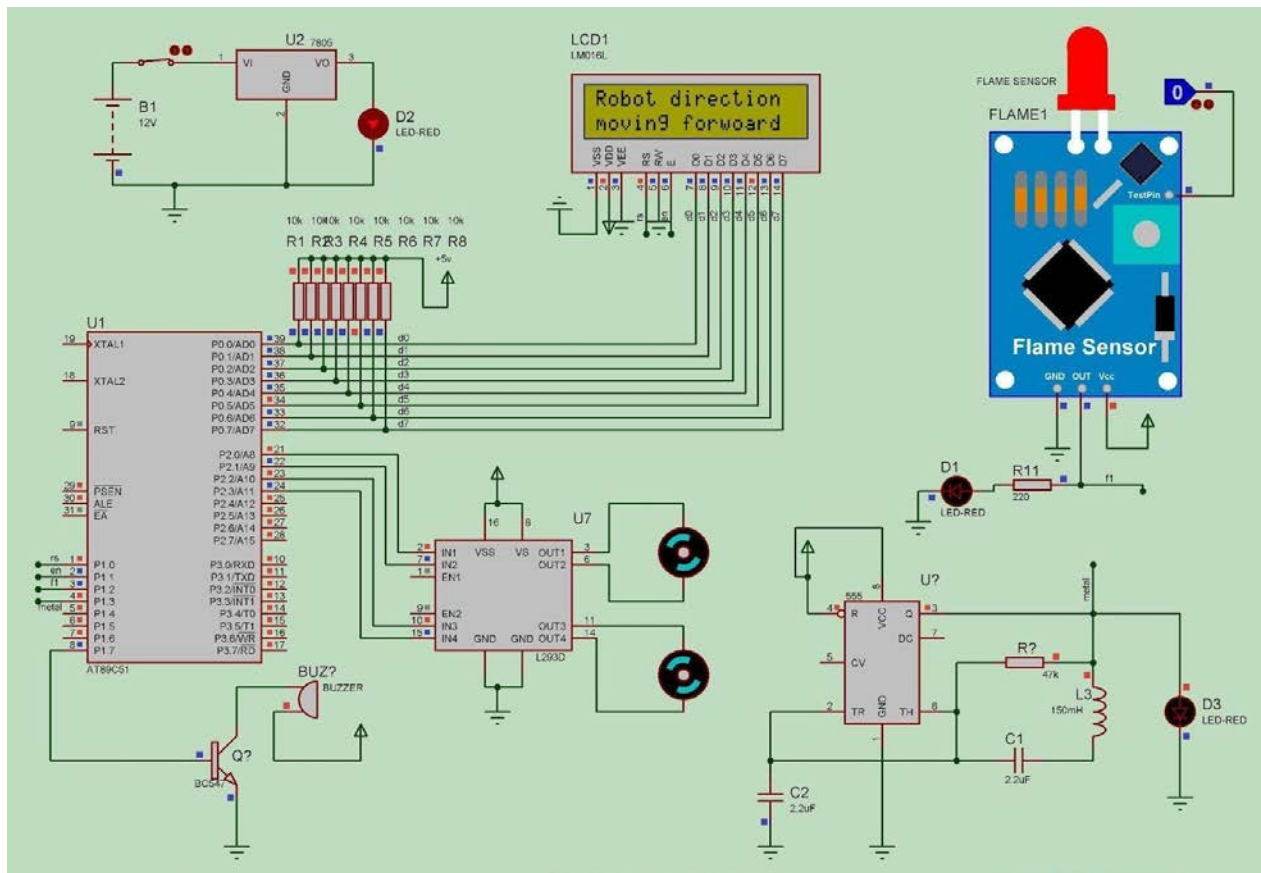


Fig. 12: Circuit simulation procedure

Furthermore the flame sensor performs its operation by detecting the fire and providing input to the microcontroller. Then microcontroller instructs the motor driver IC to stop the motor and ultimately robot movement stops whereas buzzer provides the alert sound however LCD displayed the respective status as illustrated in below Fig.-

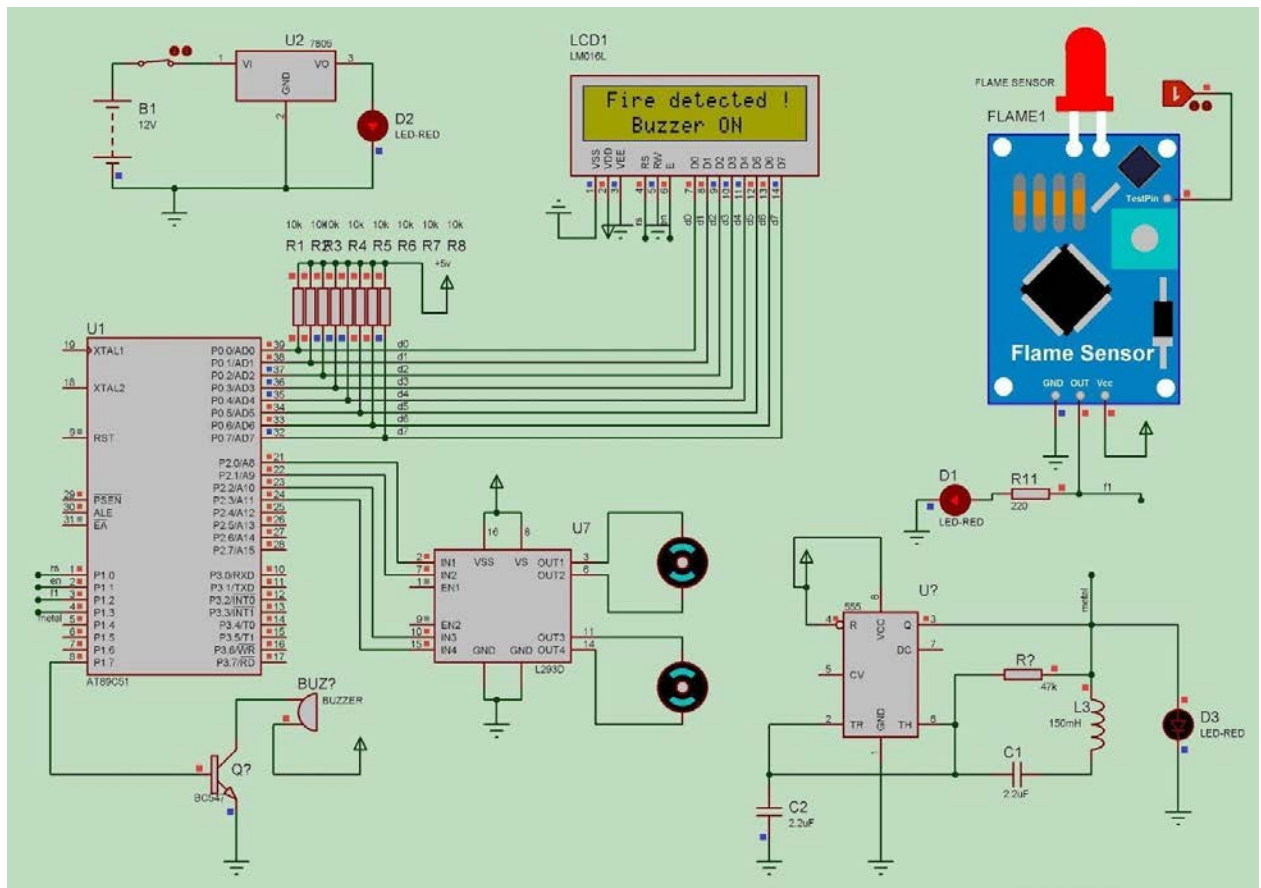


Fig. 13: Circuit simulation procedure

In case of metal detection circuit 555-IC considered and connected with $R=47K\Omega$, $L=150mH$, $C=2.2\mu F$, however this circuit configuration interfaced with microcontroller in such way that the circuit detects metal piece in the near region in terms of variation in the impedance value. Based on this microcontroller stops robot movement and alert given through buzzer and LED, whereas LCD displayed the status as given in below Fig.-

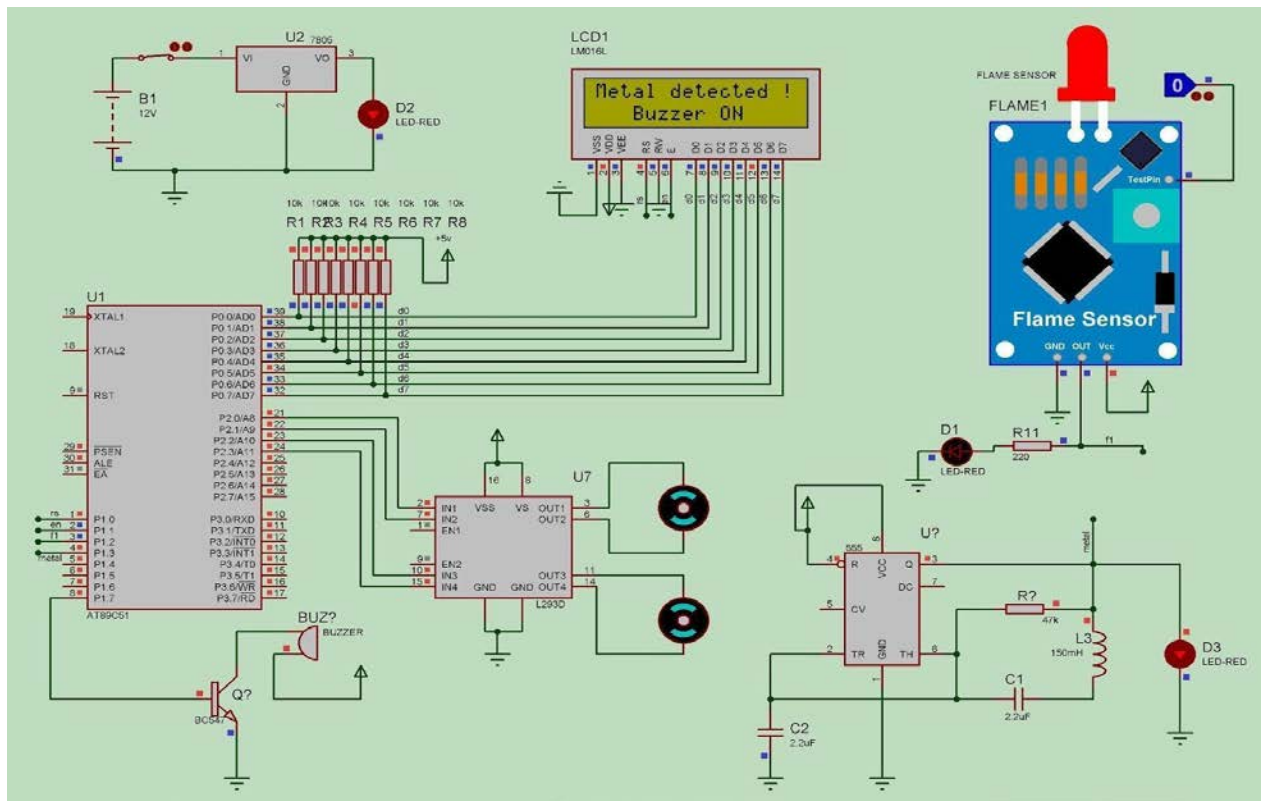


Fig. 14: Circuit simulation procedure

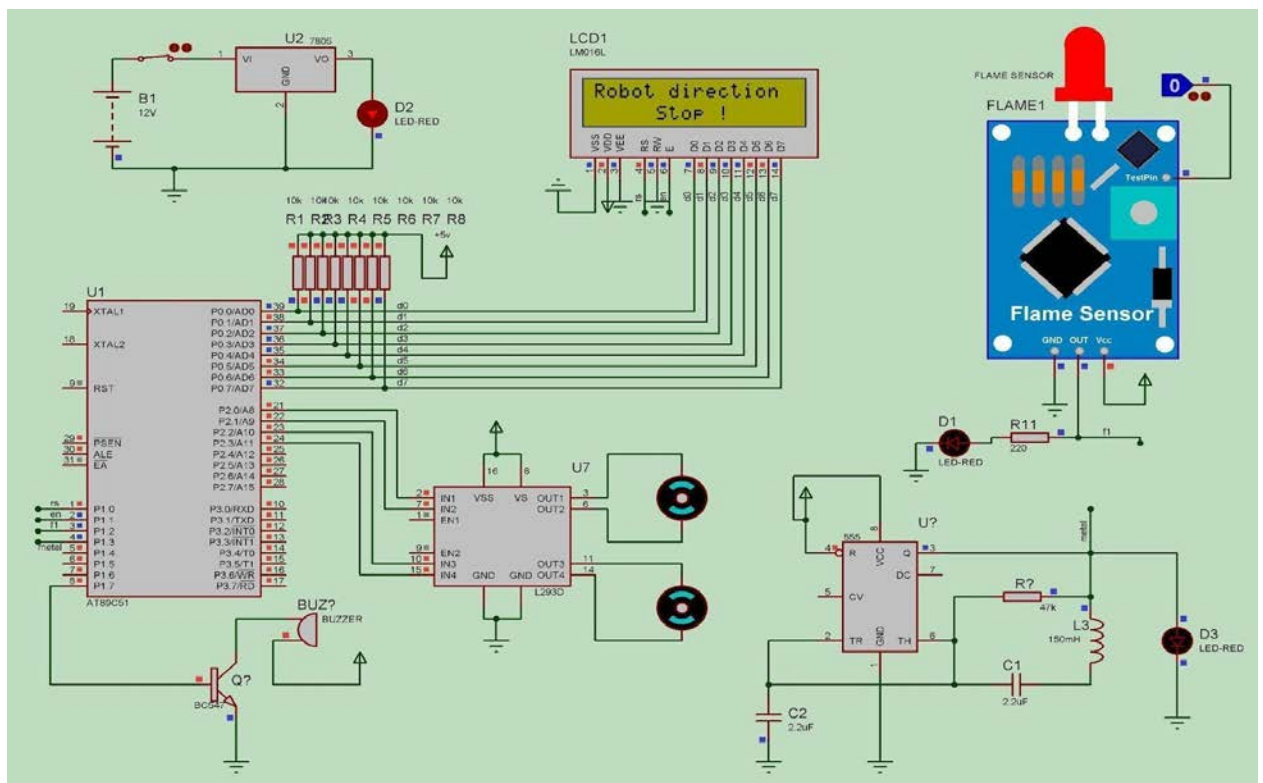


Fig. 15: Circuit simulation procedure

In the simulation, the working function of the developed robot was explained, where fire detection was carried out using a flame sensor and a 555-timer IC connected with R-L-C components to construct a metal detecting circuit, but the network elements were interfaced with the controller using the right technique. Two DC motors were used with the integration of a motor driver ICL293D and a microcontroller to drive the robot in the correct direction and for optimal activities.

The system was initially initialized, and the robot began moving forward. To confirm the detection of fire and metal, these two were integrated with LEDs that illuminate when the detection occurs, but the buzzer also activates, and the LCD shows the relevant detection. When the robot detects fire or metal, it comes to an automated halt.

The buzzer was given and combined with the controller so that the buzzer would be activated when the microcontroller delivered a signal based on the inputs received from the sensors installed, namely the fire and metal sensor, according to the planned circuit. All of the electronic parts were arranged and interfaced in the circuit in a

methodical manner, resulting in the desired robot design and framework.

It was purposefully involved in the construction of a hardware circuit model of the robot and, for this general breadboard used for installing the circuit elements during the last phase. The appropriate electricity was supplied to the breadboard, and the robot's functions were displayed. Manpower would be saved, and accuracy and mine detection efficiency would be enhanced, thanks to the devised technology. The following is the hardware circuit modeling and functioning demonstration:

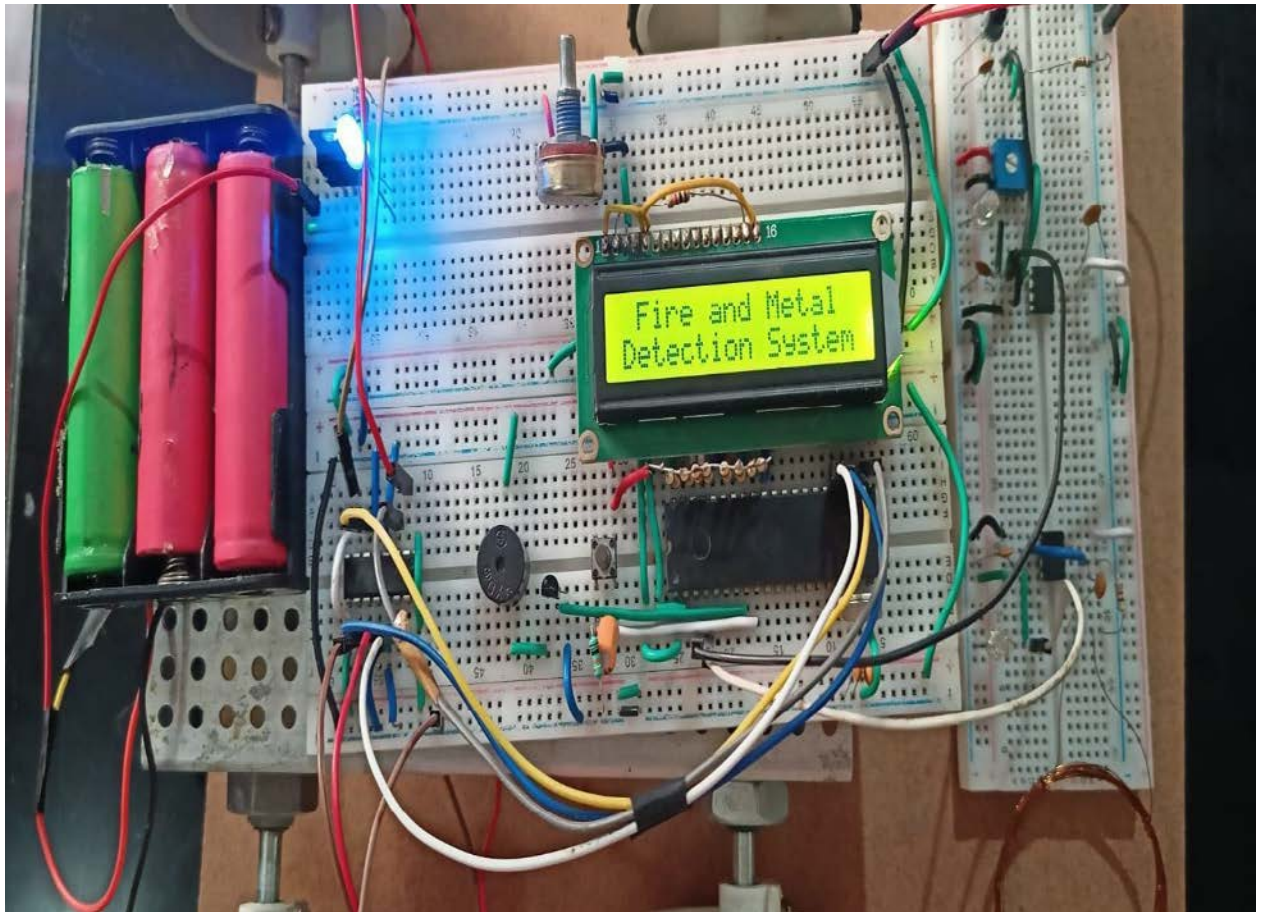


Fig. 16: Prototype Development

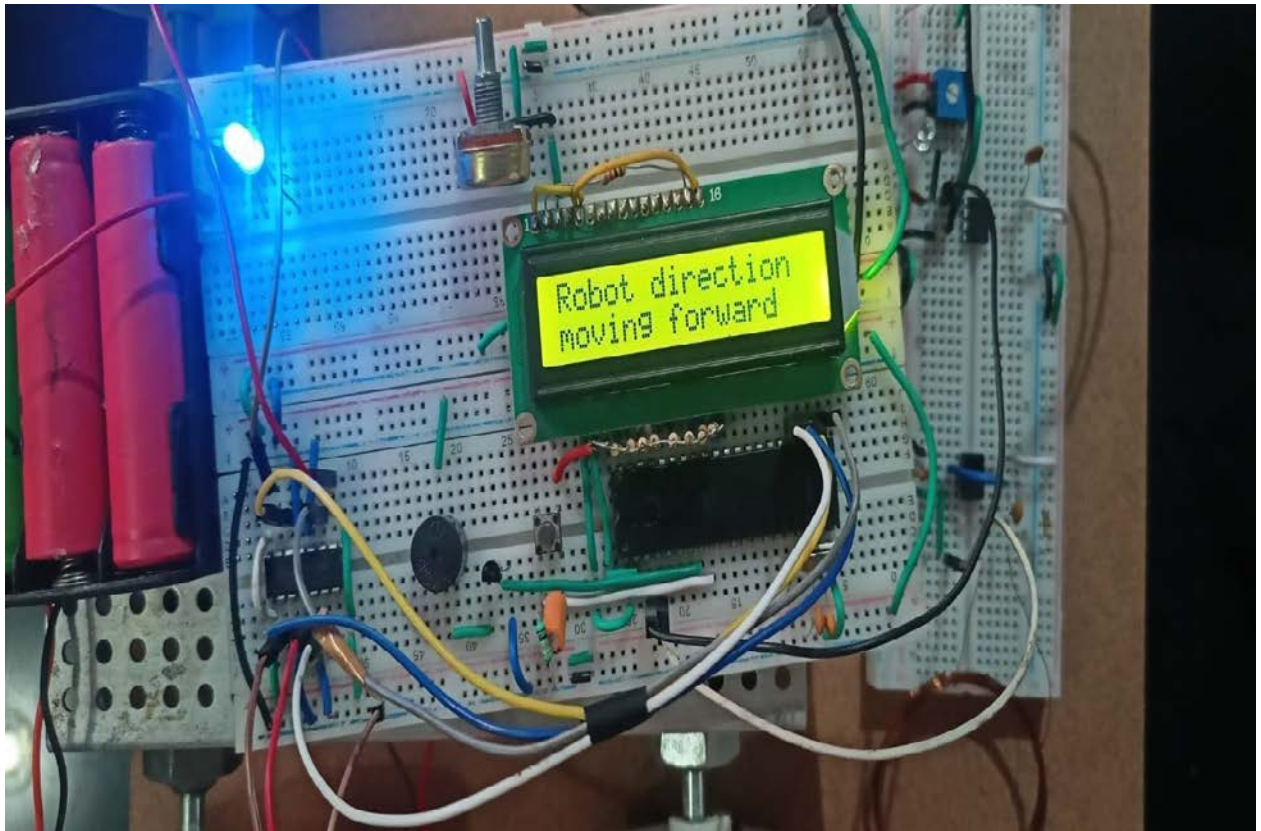


Fig. 17: Hardware prototype

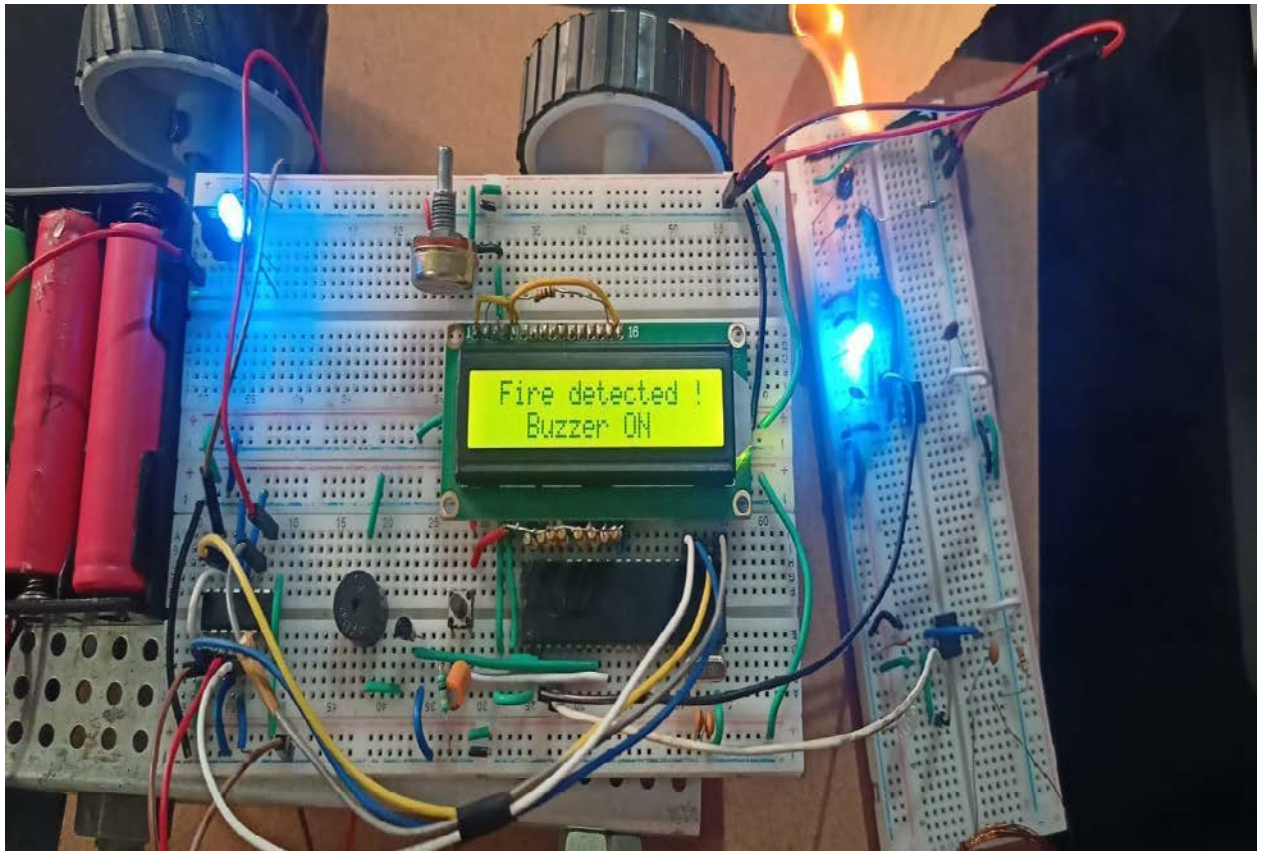


Fig. 18: Hardware prototype

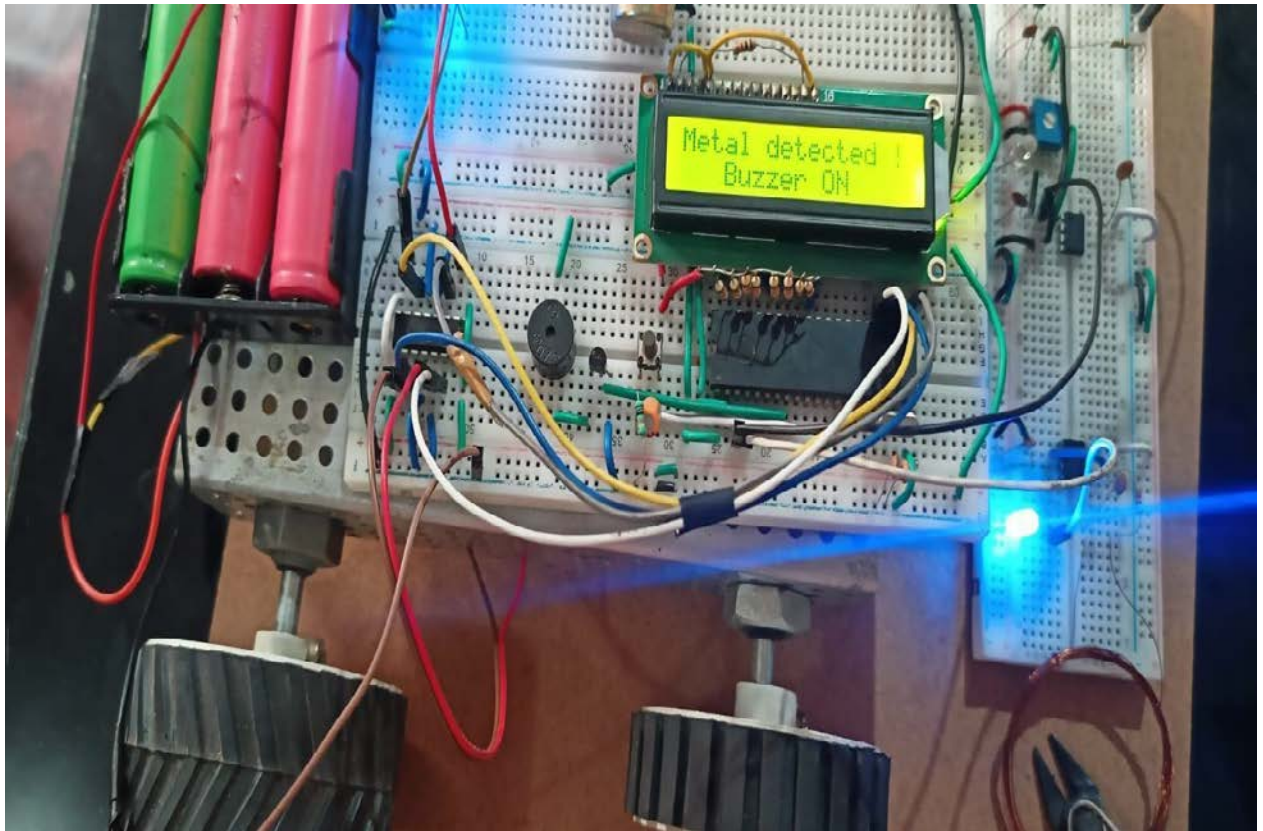


Fig. 19: Hardware prototype

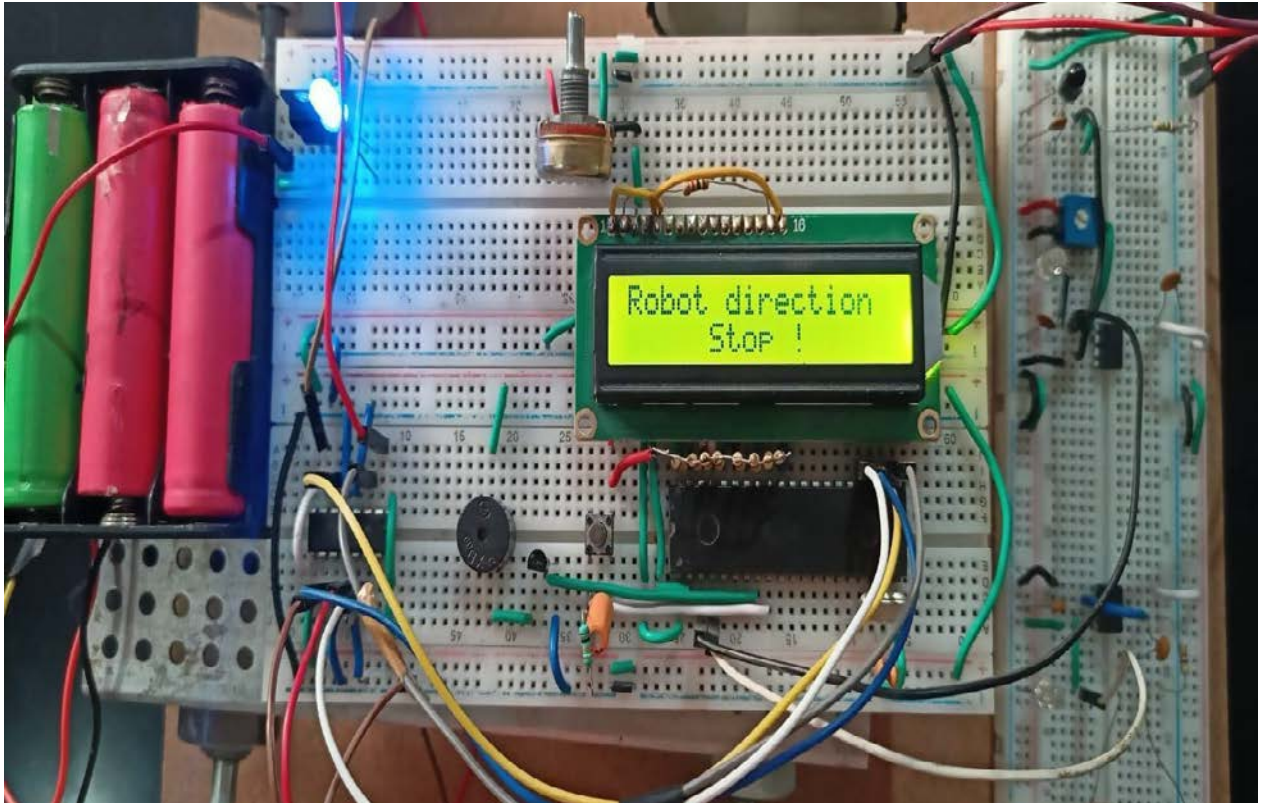


Fig. 20: Hardware prototype

The unneeded challenges encountered throughout the system design were overcome with technical understanding and electronics engineering abilities. For example, constructing a metal detecting circuit using a 555-timer was not straightforward. Thus the pin specifications of the 555-timer IC were extensively researched before the appropriate circuit connections were made.

The constructed metal detector circuit was then connected to the AT89C51 microcontroller. In this approach, the robot design assignment was completed efficiently, with the knowledge and problem-solving abilities to overcome associated complexities.

Programming expertise aided incorrect controller coding and the creation of an ideal framework for dealing with complications. Working step by step decreased the number of flaws and roadblocks significantly during the process of the study.

ANALYSIS & CONCLUSIN

The final construction work was viewed as a fantastic opportunity to increase engineering knowledge and abilities. Project-related information was gathered by researching the contents of publications and books, which supported me in comprehending the task. Various techniques and processes were taught throughout the execution of tasks, which boosted contextual knowledge.

It chose novel strategies and implemented them to complete the program's tasks. The problems that arose while working for the company The rules and standards that support the engineering institution were followed, resulting in the legal requirements being met. The work was completed precisely and within the specified deadlines.

The duties and tasks were completed with professionalism, and the work was delivered on time. An AT89C51 microcontroller was used

during this project to create a robot for fire and metal detection. It was methodically worked on, with full use of electrical engineering talents for the program's efficient completion and robot design. Working in an integrated development environment for efficient controller coding also required software capabilities.

The task was completed, and all project stages were completed in an intended way by doing the robot design-related duties step by step. The controller's activities and those of other network elements were represented using a supply unit. The elements were interfaced with the microcontroller in order to obtain the desired framework. The technique might be used to detect fire and metal mines effectively.

The precision and reliability of detecting fire and mine would be improved thanks to automated robotic activities. For the design process, the best feasible efforts were made. It made a substantial contribution to the successful completion of the task.

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